Fig. 4.4.4. A TK^5 or $TK_{3,3}$ in $G + z_1z_2$

two branch vertices of a $TK_{3,3}$. Hence K is a $TK_{3,3}$ with only one branch vertex v in, say, $G_2 - G_1$. But then also the graph $G_1 + v + \{vx, vy, vz_1\}$, which is planar by the choice of z_1 , contains a $TK_{3,3}$. This contradicts Corollary 4.2.11. $\square$

Theorem 4.4.6. (Kuratowski 1930; Wagner 1937)

The following assertions are equivalent for graphs G :

- (i) G is planar;
- (ii) G contains neither K^5 nor $K_{3,3}$ as a minor;
- (iii) G contains neither K^5 nor $K_{3,3}$ as a topological minor.

Proof. Combine Corollary 4.2.11 with Lemmas 4.4.2, 4.4.3 and 4.4.5. $\square$

[4.5.1]
[5.1.2]
[12.6.4]

(4.2.11)

Corollary 4.4.7. Every maximal planar graph with at least four vertices is 3-connected.

Proof. Apply Lemma 4.4.5 and Theorem 4.4.6. $\square$

4.5 Algebraic planarity criteria

One of the most conspicuous features of a plane graph G are its *facial cycles*, the cycles that bound a face. If G is 2-connected it is covered by its facial cycles, so in a sense these form a ‘large’ set. In fact, the set of facial cycles is large even in the sense that they generate the entire cycle space: every cycle in G is easily seen to be the sum of the facial cycles (see below). On the other hand, the facial cycles only cover G ‘thinly’, as every edge lies on at most two of them. Our first aim in this section is to show that the existence of such a large yet thinly spread family of cycles is not only a conspicuous feature of planarity but lies at its very heart: it characterizes it.

facial
cycles

Let $G = (V, E)$ be any graph. We call a subset $\mathcal{F}$ of its edge space $\mathcal{E}(G)$ *sparse* if every edge of G lies in at most two sets of $\mathcal{F}$. For example, the cut space $\mathcal{B}(G)$ has a sparse basis consisting of atomic cuts (Proposition 1.9.2).

sparse

[4.6.3]

Theorem 4.5.1. (MacLane 1937)*A graph is planar if and only if its cycle space has a sparse basis.*(1.9.1)
(1.9.5)
(4.1.1)
(4.2.2)
(4.2.6)
(4.4.6)

Proof. The assertion being trivial for graphs of order at most 2, we consider a graph G of order at least 3. If $\kappa(G) \leq 1$, then G is the union of two proper induced subgraphs G_1, G_2 with $|G_1 \cap G_2| \leq 1$. Then $\mathcal{C}(G)$ is the direct sum of $\mathcal{C}(G_1)$ and $\mathcal{C}(G_2)$, and hence has a sparse basis if and only if both $\mathcal{C}(G_1)$ and $\mathcal{C}(G_2)$ do (proof?). Moreover, G is planar if and only if both G_1 and G_2 are: this follows at once from Kuratowski's theorem, but also from easy geometrical considerations. The assertion for G thus follows inductively from those for G_1 and G_2 . For the rest of the proof, we now assume that G is 2-connected.

We first assume that G is planar and choose a drawing. By Proposition 4.2.6, the face boundaries of G are cycles, so they are elements of $\mathcal{C}(G)$. We shall show that the face boundaries generate all the cycles in G ; then $\mathcal{C}(G)$ has a sparse basis by Lemma 4.2.2. Let $C \subseteq G$ be any cycle, and let f be its inner face. By Lemma 4.2.2, every edge e with $\bar{e} \subseteq f$ lies on exactly two face boundaries $G[f']$ with $f' \subseteq f$, and every edge of C lies on exactly one such face boundary. Hence the sum in $\mathcal{C}(G)$ of all those face boundaries is exactly C .

Conversely, let $\{C_1, \dots, C_k\}$ be a sparse basis of $\mathcal{C}(G)$. Then, for every edge $e \in G$, also $\mathcal{C}(G - e)$ has a sparse basis. Indeed, if e lies in just one of the sets C_i , say in C_1 , then $\{C_2, \dots, C_k\}$ is a sparse basis of $\mathcal{C}(G - e)$; if e lies in two of the C_i , say in C_1 and C_2 , then $\{C_1 + C_2, C_3, \dots, C_k\}$ is such a basis. (Note that $C_1 + C_2$ is indeed an element of $\mathcal{C}(G - e)$, by Proposition 1.9.1.) Thus every subgraph of G has a cycle space with a sparse basis. For our proof that G is planar, it thus suffices to show that the cycle spaces of K^5 and $K_{3,3}$ (and hence those of their subdivisions) do *not* have a sparse basis: then G cannot contain a TK^5 or $TK_{3,3}$, and so is planar by Kuratowski's theorem.

Let us consider K^5 first. By Theorem 1.9.5, $\dim \mathcal{C}(K^5) = 6$; let $\mathcal{B} = \{C_1, \dots, C_6\}$ be a sparse basis, and put $C_0 := C_1 + \dots + C_6$. As $\mathcal{B}$ is linearly independent, none of the sets $C_0, \dots, C_6$ is empty, so each of them contains at least three edges (cf. Proposition 1.9.1). Moreover, as every edge from C_0 lies in just one of $C_1, \dots, C_6$, the set $\{C_0, \dots, C_6\}$ is still sparse. But this implies that K^5 should have more edges than it does, i.e. we obtain the contradiction of

$$21 = 7 \cdot 3 \leq |C_0| + \dots + |C_6| \leq 2 \|K^5\| = 20.$$

For $K_{3,3}$, Theorem 1.9.5 gives $\dim \mathcal{C}(K_{3,3}) = 4$; let $\mathcal{B} = \{C_1, \dots, C_4\}$ be a sparse basis, and put $C_0 := C_1 + \dots + C_4$. As $K_{3,3}$ has girth 4, each C_i contains at least four edges. We then obtain the contradiction of

$$20 = 5 \cdot 4 \leq |C_0| + \dots + |C_4| \leq 2 \|K_{3,3}\| = 18.$$

□

A constructive proof of the backward implication of MacLane's theorem is indicated in Exercise 32. That proof shows that the generating set we chose in our proof of the forward implication, the set of face boundaries, is canonical in the following sense: given any set $\mathcal{D}$ of cycles in a 2-connected planar graph G that generates $\mathcal{C}(G)$ and is such that every edge of G lies on exactly two of those cycles, there is a drawing of G in which the cycles in $\mathcal{D}$ are precisely the face boundaries.

It is one of the hidden beauties of planarity theory that two such abstract and seemingly unintuitive results about generating sets in cycle spaces as MacLane's theorem and Tutte's theorem 3.2.6 conspire to produce a very tangible planarity criterion for 3-connected graphs:

Theorem 4.5.2. (Kelmans 1978)

A 3-connected graph is planar if and only if every edge lies on at most (equivalently: exactly) two non-separating induced cycles.

(3.2.6)

Proof. The forward implication follows from Proposition 4.2.7 and Lemma 4.2.2 (and Proposition 4.2.6 for the 'exactly two' version); the backward implication follows from Theorems 3.2.6 and 4.5.1. $\square$

(4.2.2)

(4.2.6)

(4.2.7)

Let us conclude this section with another characterization of planarity, one with a very different flavour. A *linear extension* of a partial ordering $\leq$ of a set P is a total ordering $\leq'$ on P which includes $\leq$ as a subset of P^2 . Thus, for any $p \leq q$ in P we still have $p \leq' q$, and for incomparable $p, q \in P$ we have either $p <' q$ or $p >' q$ in addition. The *dimension* of the partially ordered set $(P, \leq)$ is the least number of linear extensions of $\leq$ on P whose intersection is exactly $\leq$: for any incomparable $p, q \in P$ there must be a linear extension $\leq'$ with $p <' q$ and another linear extension $\leq''$ with $p >' q$ in this collection.

linear
extension

poset
dimension

With every graph $G = (V, E)$ one can associate its *incidence poset*, the partially ordered set $(V \cup E, \leq)$ in which $v < e$ if and only if v is a vertex and e is an edge at v . (Thus, as a relation, $<$ is the same as $\in$.)

incidence
poset

Theorem 4.5.3. (Schnyder 1989)

A graph is planar if and only if its incidence poset has dimension ≤ 3 .

4.6 Plane duality

In this section we shall use MacLane's theorem to uncover another connection between planarity and algebraic structure: a connection between the duality of plane graphs, defined below, and the duality of the cycle and cut space hinted at in Chapters 1.9 and 2.4.

plane
multigraph

A *plane multigraph* is a pair $G = (V, E)$ of finite sets (of *vertices* and *edges*, respectively) satisfying the following conditions:

- (i) $V \subseteq \mathbb{R}^2$;
- (ii) every edge is either an arc between two vertices or a polygon containing exactly one vertex (its *endpoint*);
- (iii) apart from its own endpoint(s), an edge contains no vertex and no point of any other edge.

We shall use terms defined for plane graphs freely for plane multigraphs. Note that, as in abstract multigraphs, both loops and double edges count as cycles.

Let us consider the plane multigraph G shown in Figure 4.6.1. Let us place a new vertex inside each face of G and link these new vertices up to form another plane multigraph G^* , as follows: for every edge e of G we link the two new vertices in the faces incident with e by an edge e^* crossing e ; if e is incident with only one face, we attach a loop e^* to the new vertex in that face, again crossing the edge e . The plane multigraph G^* formed in this way is then dual to G in the following sense: if we apply the same procedure as above to G^* , we obtain a plane multigraph very similar to G ; in fact, G itself may be reobtained from G^* in this way.

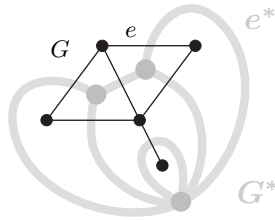


Fig. 4.6.1. A plane graph and its dual

To make this idea more precise, let $G = (V, E)$ and (V^*, E^*) be any two plane multigraphs, and put $F(G) =: F$ and $F((V^*, E^*)) =: F^*$. We call (V^*, E^*) a *plane dual* of G , and write $(V^*, E^*) =: G^*$, if there are bijections

plane dual
 G^*

$$\begin{array}{lll} F \rightarrow V^* & E \rightarrow E^* & V \rightarrow F^* \\ f \mapsto v^*(f) & e \mapsto e^* & v \mapsto f^*(v) \end{array}$$

satisfying the following conditions: